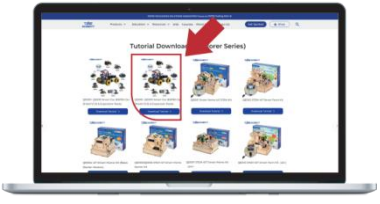
ACEBOTT

Smart Car Manual

Important Note:

1. This manual only contains the installation steps and control method of this product. If you need to know more control methods and detailed tutorials, please visit the official website to download the relevant tutorials:
https://acebott.com/tutorial/
QD001-QD010 Smart Car (ESP32-Car-Shield V2.0) &Expansion Packs



2. Due to shipping policy prohibiting the shipment of 18650 batteries with products, this kit does "NOT INCLUDE 18650 BATTERIES".
To purchase the required"BUTTON-TOP" 18650 rechargeable batteries separately, please scan/refer to the QR code below.
Thankyou foryour understanding!^ _^



Scan the QR code above to get the purchase link
This product needs tobe equipped with 18650 BUTTON TOP TYPE Battery *2

3. Warranty Service
Free maintenance: ✓ Electronic components: 3 months ✓ Circuit board: 1 year
Email us at: **support@acebott.com**

YouTube | Facebook | Instagram | Tik Tok

Follow us for more product info!

I. Safety Instructions

1. General Safety

1.Attention! This kit is not suitable for children under 3 years old due to small parts that may cause a choking hazard.
2.This product is designed as an educational kit for children aged 10 and above.
3.Assembly must be carried out under adult supervision.
4.Sharp edges or pointed parts may cause injuries. Please handle with care.
5.Never manually rotate the motor, servo, or other movable parts—there is a risk of damage or personal injury.

2. Electrical Safety

1.Use only protected 18650 lithium-ion batteries (3.7V) with integrated protection circuit (PCM).
2.Do not use unprotected or damaged batteries, as they may pose serious safety risks.
3.Pay attention to the correct polarity when inserting the battery.
4.Use only a suitable charger to avoid overcharging, short-circuiting, or overheating.
5.The vehicle must be operated under supervision. Always switch off the power after use.
6.If the device will not be used for a long time, remove the battery to prevent deep discharge or corrosion.

II. Cleaning and Maintenance

1.Clean the device only with a dry, lint-free cloth.
2.Do not use cleaning agents, solvents, or water.
3.Regularly check the condition of the battery and electronic components.
4.If the battery shows visible damage, stop using it immediately.

Product List



ESP32 Max V1.0 Controller Board
1PC



ESP32-Car-Shield V2.0
1PC



Blue LED Module
2PCS



P-Buzzer Module
1PC



IR Receiver Module
1PC



Ultrasonic Sensor
1PC



Trace Sensor
1PC



TT DC Geared Motor
4PCS



Mecanum Wheel
4PCS



IR Romote Control
1PC



Servo SG90 9G
1PC



18650 Battery Holder
1PC



Motor Bracket
4PCS



Coupler
4PCS



USB Cable
1M
1PC



F-F 3P Dupont Wire
6PCS



F-F 4P Dupont Wire
2PCS



F-F 5P Dupont Wire
2PCS



Screwdriver
1PC



Cross Wrench
1PC



Ultrasonic Bracket
1PC



Acrylic Board
1PC



M3*10MM Round Head Screws
39PCS



M3*30MM Round
10PCS
Head Screws



M2*10MM Round Head Screws
8PCS



M2*25MM Round Head Self-Tapping Screws
5PCS



M3*8MM Flat Head Screws
5PCS



M1.4*5 Large Round Flat Head Tapping Screws
3PCS



M2 Nickel-Plated Nuts
8PCS



M3 Nickel-Plated Nuts
31PCS



M3*14MM Single-pass Copper Pillar
5PCS



M3*40MM Dual-pass Copper Pillar
8PCS



Nylon Cable Ties
3PCS

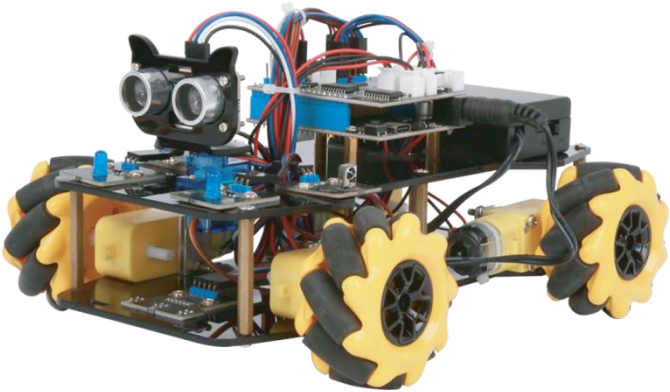


Black Tape
1PC

III. Assembly Video

Tip: Scan the QR code to watch the Assembly Video.

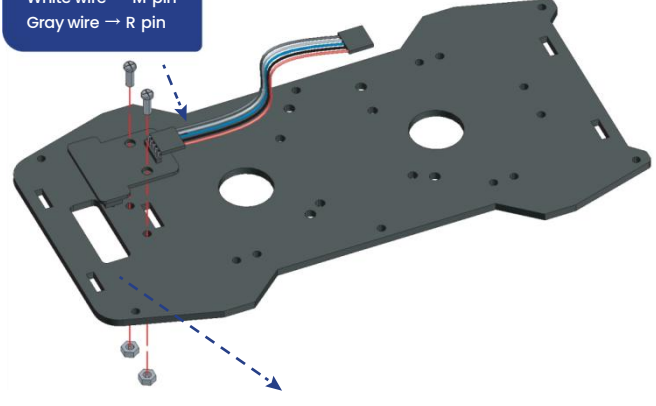
IV. Assembly Steps for The Smart Car



1. Installing the trace sensor.

Parts	
Components	**Quantity**
Acrylic Bottom Chassis	1
Trace Sensor	1
M3*10MM Round Head Screw	2
M3 Nut	2
5P Dupont Wire	1

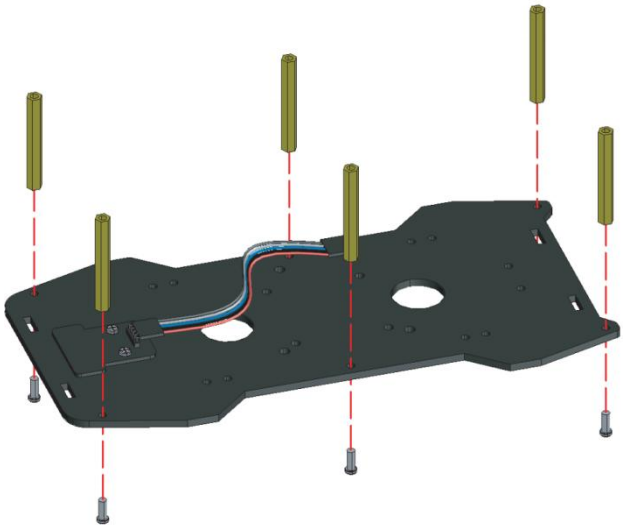
Red wire → V pin
Black wire → G pin
Blue wire → L pin
White wire → M pin
Gray wire → R pin



The bottom chassis has mounting a hole for the trace sensor.

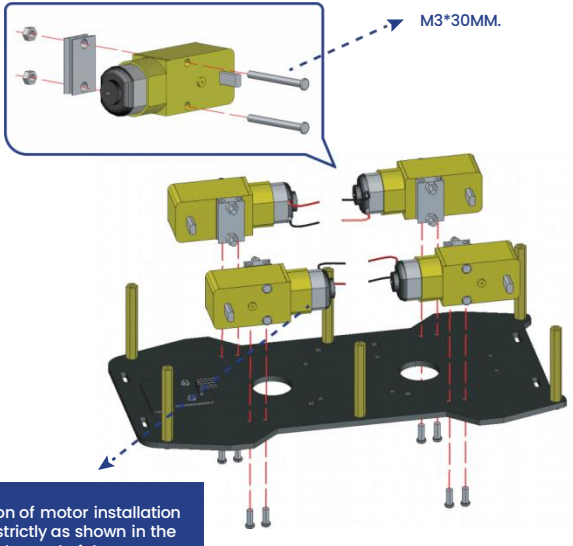
2. Installing copper pillars on the acrylic bottom chassis.

Parts	
Components	Quantity
M3*40MM Dual-pass Copper Pillar	6
M3*10MM Round Head Screw	6



3. Installing motors.

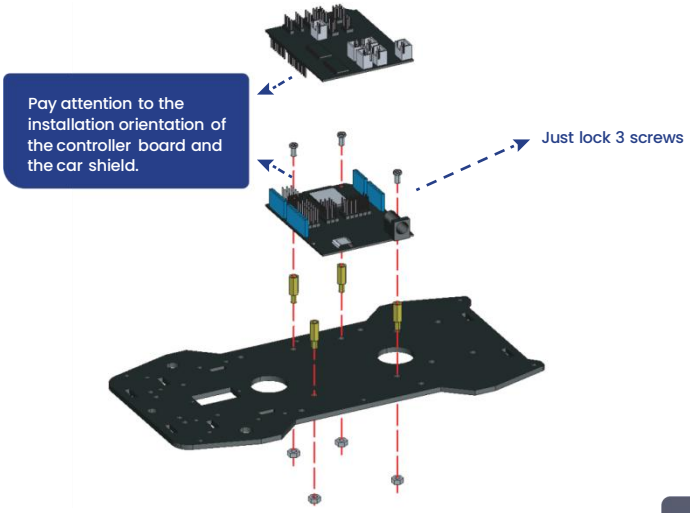
Parts	
Components	**Quantity**
TT DC Geared Motor	4
Motor Bracket	4
M3*10MM Round Head Screw	8
M3*30MM Round Head Screw	8
M3 Nut	8



The direction of motor installation should be strictly as shown in the figure, and the end of the motor belt line should be placed inward.

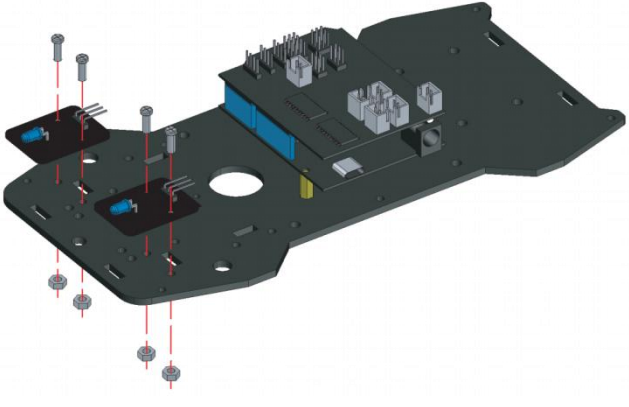
4. Installing the controller board and the car shield.

Parts	
Components	**Quantity**
ESP32 Max Controller Board	1
ESP32-Car-Shield	1
Acrylic Top Chassis	1
M3*14MM Single-pass Copper Pillar	4
M3*10MM Round Head Screw	3
M3 Nut	4



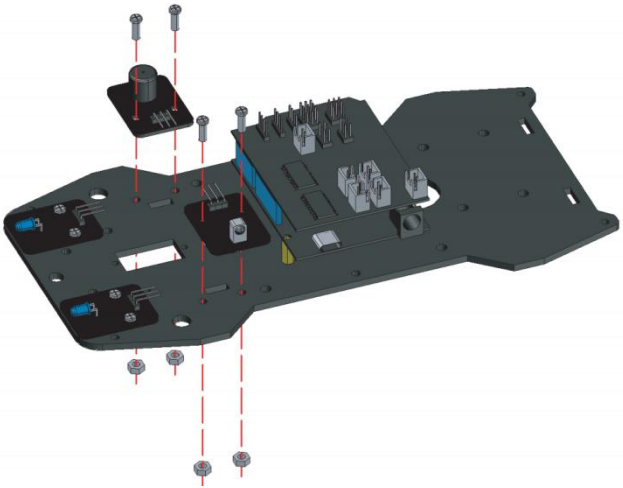
5. Installing blue LED modules.

Parts	
Components	**Quantity**
Blue LED Module	2
M3*10MM Round Head Screw	4
M3 Nut	4



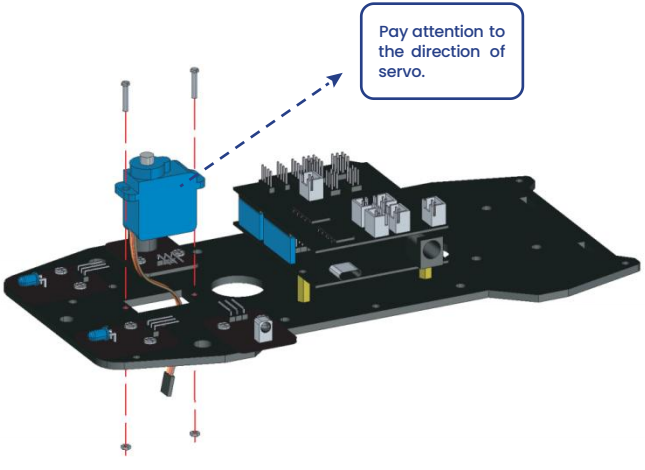
6. Installing the P-buzzer module and the IR receiver module.

Parts	
Components	Quantity
P-Buzzer Module	1
IR Receiver Module	1
M3*10MM Round Head Screws	4
M3 Nut	4



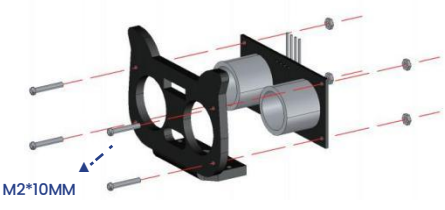
7. Installing the servo motor.

Parts	
Components	**Quantity**
Servo SG90 9G	1
M2*10MM Round Head Screws	2
M2 Nut	2

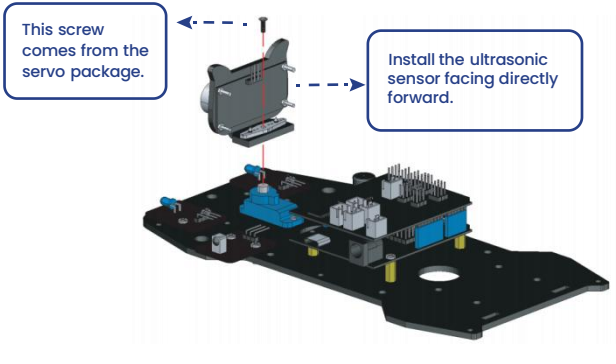


8. Installing the ultrasonic sensor.

Parts	
Components	**Quantity**
Ultrasonic Bracket	1
Ultrasonic Sensor	1
M2*10MM Round Head Screws	4
M2 Nut	4
Long Servo Horn	1
M1.4*5MM Large Round Flat Head Tapping Screws	2

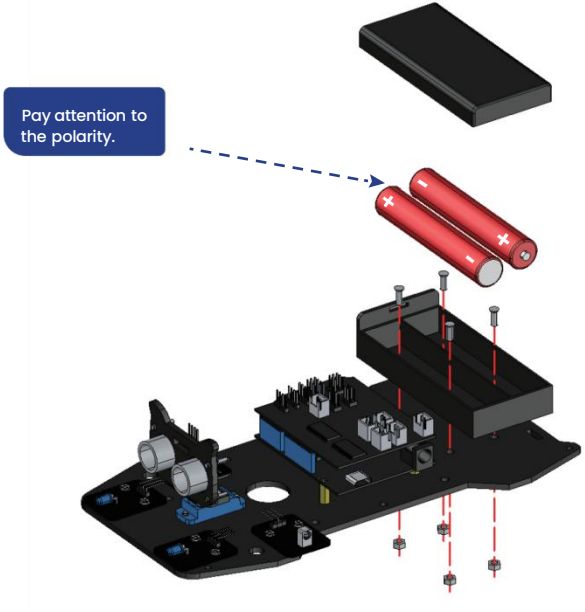






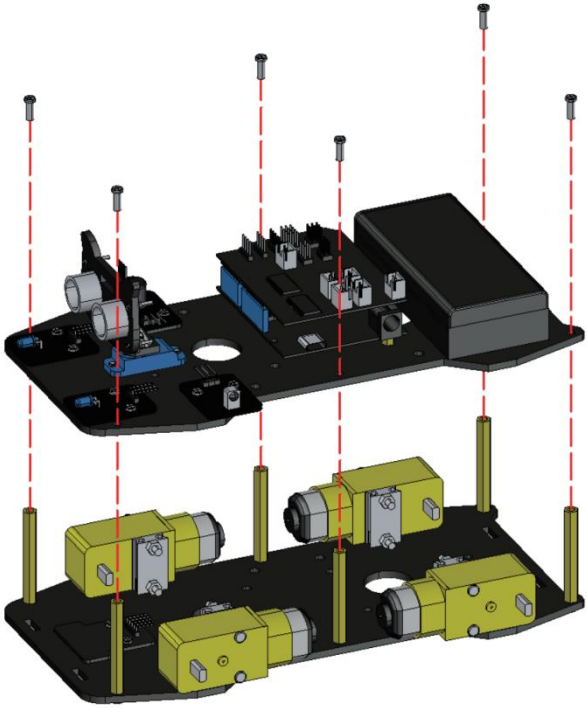
9. Installing the battery holder.

Parts	
Components	**Quantity**
18650 Battery Holder	1
18650 Battery (Self-provided)	2
M3*8MM Flat Head Screw	4
M3 Nut	4



10. Connect the bottom chassis and the top chassis.

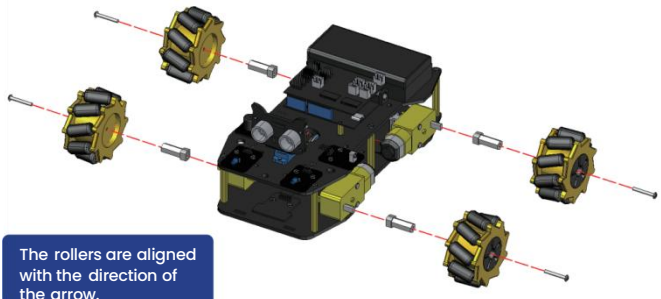
Parts	
Components	Quantity
M3*10MM Round Head Screw	6



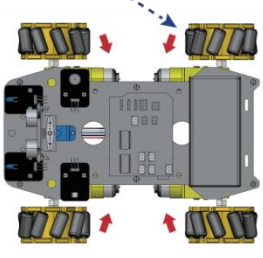
11. Installing Mecanum wheels.

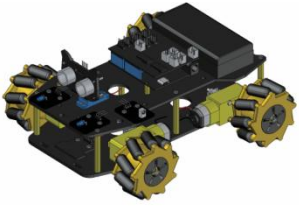
Parts	
Components	**Quantity**
60MM Mecanum Wheel	4
Mecanum Wheel Coupling	4
M2*25MM Round Head Self-Tapping Screw	4

Notes:The Mecanum wheel cannot be installed arbitrarily and needs to be installed strictly in the direction shown in the figure.



The rollers are aligned with the direction of the arrow.





12. Wiring.

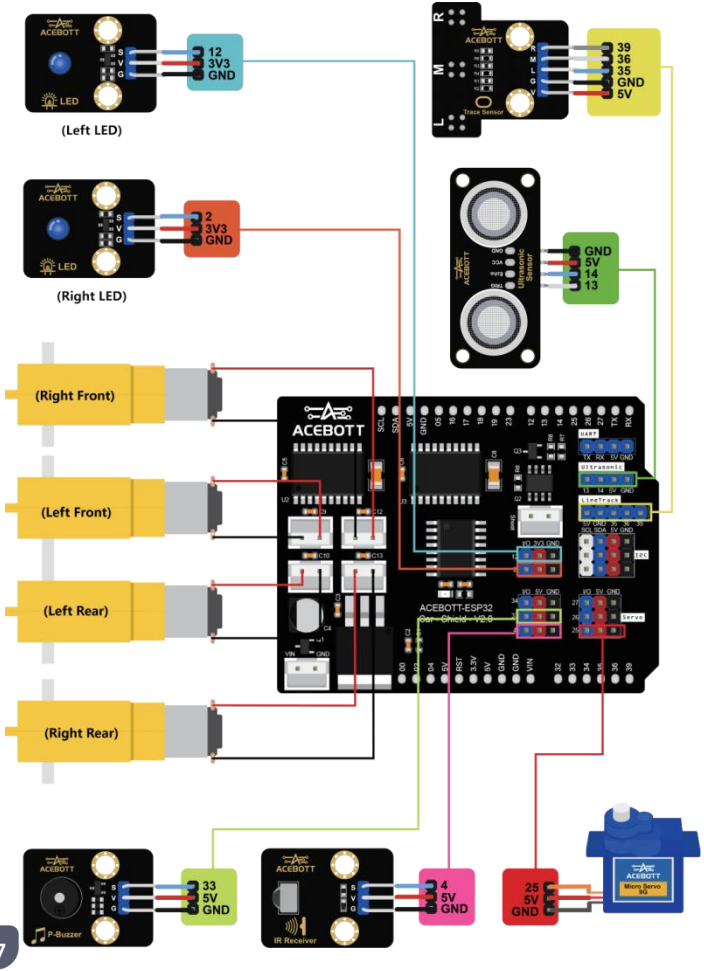
1. The DuPont wires are blue, red, and black. The blue wire connects to the S pin, the red wire to the V pin, and the black wire to the G pin.

2. The servo wire colors differ from standard DuPont wires: the red wire connects to the V pin, the brown wire to the G pin, and the yellow wire to the pin25.

3. For the ultrasonic sensor wires: the black wire connects to the GND pin, the red wire to the V pin, the blue wire to the ECHO pin (pin14), and the white wire to the TRIG pin (pin13).

4. For the line-following sensor wires: the red wire connects to the 5V pin, the black wire to the GND pin, the blue wire to the L pin (pin35), the white wire to the M pin (pin36), and the gray wire to the R pin (pin39).

5. Be sure to connect the modules to the ESP32 control board strictly according to the wiring instructions. Incorrect wiring may cause a short circuit in the ESP32 control board and result in damage.


ACEBOTT
LED
S
V
G
12
3V3
GND
(Left LED)
2
3V3
GND
(Right LED)
R
M
L
Trace Sensor
39
36
35
GND
5V
Ultrasonic Sensor
GND
5V
14
13
(Right Front)
(Left Front)
(Left Rear)
(Right Rear)
ACEBOTT-ESP32
Car Shield V2.0
UART
Ultrasonic
LineTrack
I2C
Servo
33
5V
GND
P-Buzzer
4
5V
GND
IR Receiver
25
5V
GND
Micro Servo
IO

V. APP Control of The Smart Car

1. Install the ACEBOTT APP.

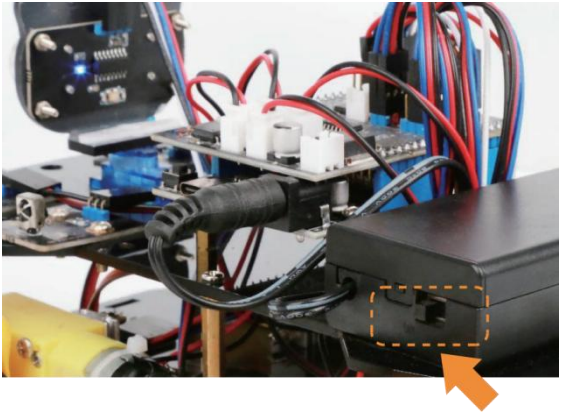


Search for "ACEBOTT" in your phone's app store, find the app, and download and install it.

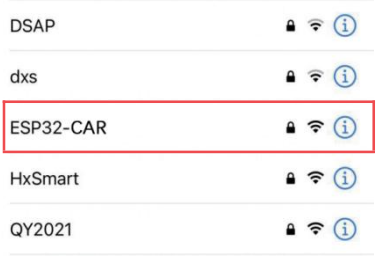
For IOS system, you need to search the keyword: ACEBOTT in the APP Store and then download it.

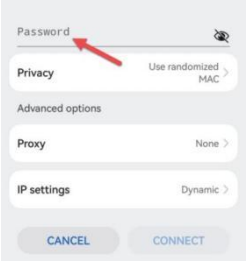
For Android system, you need to search the keyword: ACEBOTT in the Google Play and then download.

2. Take out the assembled Smart Car and flip the switch to the right to turn on the Smart Car's power.



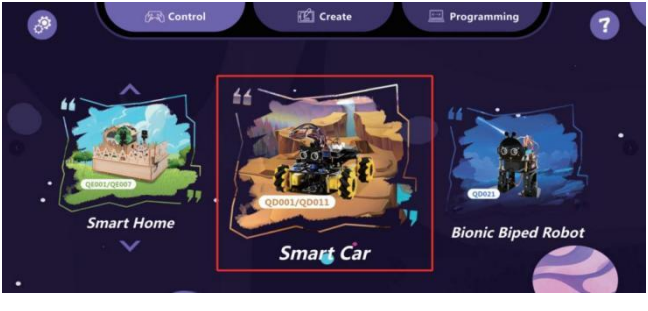
3. Open your phone's WiFi settings, search for the Smart Car's WiFi, and connect. The WiFi name should be "ESP32-CAR". The WiFi password is 12345678.





Note: If you do not find "ESP32-CAR" WiFi, please click here to go to the program burn tutorial.

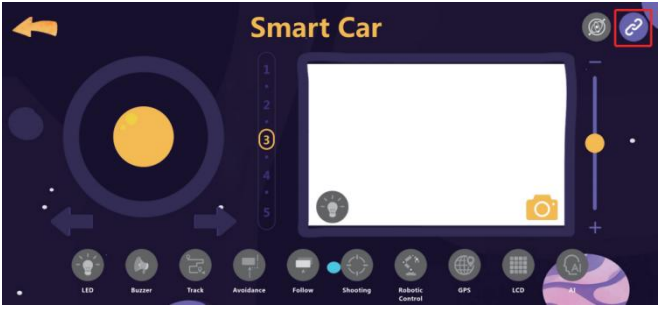
4. After connecting to WiFi, return to the app, find the "Smart Car" option, and click to enter.



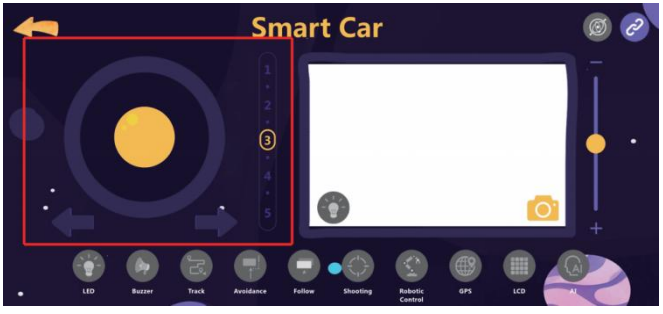
5. Tap "Control" to enter the Smart Car control interface.



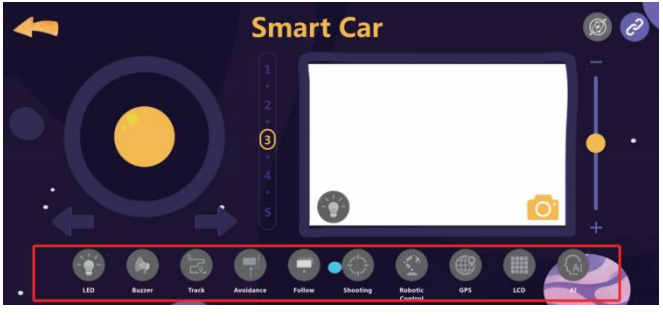
6. Now enter the control interface and click the connection icon in the upper right corner. If the icon is lit, the connection is successful.



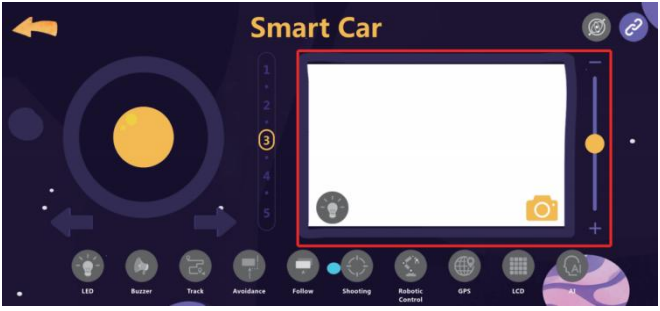
7. After the connection is successful, you can control the Smart Car. The left panel controls the basic movements of the car: forward, backward, left move, right move, turn left, turn right; the middle numbers are used for speed adjustment of the car, the larger the number, the faster the speed.



8. Below is a row of buttons for controlling the functions of the car, from left to right, they control the following functions of the car: LED function, music playback, track mode, avoidance mode, follow mode, shooting(QD005 expansion pack), robotic arm(QD007 expansion pack), GPS(QD009 expansion pack), AI(QD003 expansion pack) function.



9. The central control panel is for the car's camera function(QD002 expansion pack), and the slider on the right is for servo control. Both of these features are extended functions of the car and require the corresponding expansion packs to be used.



10. In the upper right corner of the smart car's operation interface, there is a gyroscope control provided. After clicking this button, you can control the movement of the car through the gyroscope of your mobile phone.



